

ISWE – Tutorial 9

CASE Tools

1. What is CASE tool?

A CASE (Computer Aided Software Engineering) tool is software that helps developers design, develop, test, and maintain software systems efficiently by automating parts of the software development process.

2. Five categories of CASE tools

- Upper CASE tools
- Lower CASE tools
- Integrated CASE tools
- Cross life cycle tools
- Supporting CASE tools

3. Upper CASE Tools

Upper CASE tools are used in the early stages of software development, mainly during system analysis and design.

Functions:

- Requirement analysis
- System modeling (DFD, ERD, UML)
- Database design
- Documentation support

4. Lower CASE Tools

Lower CASE tools are used in the later stages of development such as coding, testing, and maintenance.

Functions:

- Code generation
- Debugging

- Testing
- System maintenance

5. CASE Tools Description

a. Cross life cycle tools

These tools support multiple phases of the software development life cycle.

Examples:

- Project management
- Configuration management
- Documentation management

b. Code generator

A code generator automatically converts design models into source code, reducing manual coding effort and errors.

c. Reverse engineering tools

Reverse engineering tools analyze existing software to recover design, structure, or documentation from source code.